Bivariate data with technology

1

a $y = -0.29x + 0.67$

b $y = 0.12x - 1.75$

2

a

i -0.47

ii Weak negative linear correlation

b

i -0.84

ii Strong negative linear correlation

c

i 0.47

ii Weak positive linear correlation

d

i -0.91

ii Strong negative linear correlation

e

i -0.59

ii Moderate negative linear correlation

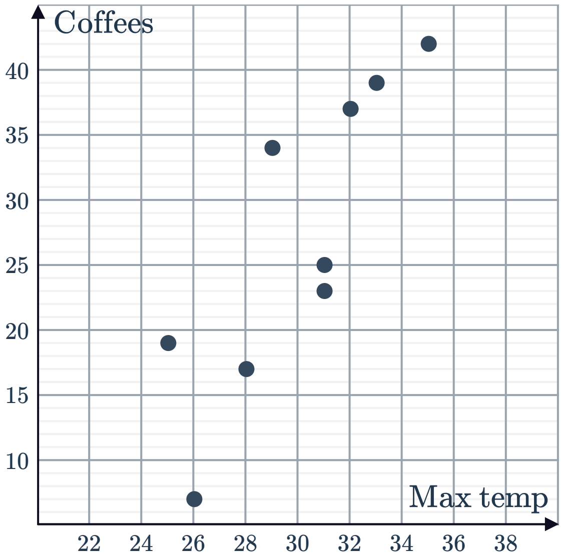
f

i 0.19

ii Weak positive linear correlation

3

a



b $r = 0.84$

c As the temperature increases, the sales tend to increase.

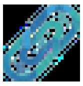
4

a -0.85

b Strong negative linear relationship

c $y = -0.1x + 30.3$

5

- **a** 0.60
-  **b** Moderate positive linear relationship
- **c** $y = 4.2x - 34.5$

6

- **a** $y = -1694x + 18\,407$
- **b** $18\,407$
- **c** $\$18\,407$
- **d** -1694
- **e** The gradient of the line indicates that age and price have a negative correlation.
- **f** The price decreases by $\$1694$.

Predictions

7

- **a** $r = -0.908$
- **b** Strong negative linear relationship
- **c** $y = -0.06x + 6.37$
- **d** 6.07 kg
- **e** Reliable due to interpolation and a strong correlation.

8

- **a** $r = 0.997$
- **b** Strong positive linear relationship
- **c** $y = 0.85x + 1.28$
- **d** 52.28%
- **e** Reliable due to interpolation and a strong correlation.

9

- **a** $V = 14.9P + 30.3$
- **b** 30.3
- **c** 30.3 words
- **d** Yes, as children learn words in other ways.
- **e** 14.9
- **f** For each additional page that a child reads each day, the child's vocabulary will increase by 14.9 words.

10

a $P = -0.2M + 99.6$

b 99.6

c 99.6%

d Yes, it is reasonable because they will have more time to study.

e -0.2

f For each additional minute that a student uses SnappyChatty, their test mark will decrease by 0.2%.

11

a $r = 0.79$

b Strong positive linear relationship

c $y = 0.025x - 550.108$

d $\$1449.89$

e $\$80\,000$ because the other two are examples of extrapolation.

12

a $r = 0.86$

b Strong positive linear relationship

c $y = 0.89x + 2.78$

d $25.03^{\circ}C$

e $32^{\circ}\mathrm{C}$, because the other two values are examples of extrapolation.

13

a $r = -0.91$

b Strong negative linear relationship

c $y = -6.6x + 80.1$

d 47.1

e Reliable due to interpolation and a strong correlation.

14

a 0.72

b Strong positive linear relationship

c $A = 1.61E - 16.63$

d 39.72%

e **C**

15

- **a** 0.86
- **b** Strong positive linear relationship
- **c** $y = 0.98x - 7.94$
- **d** $\$50.86$
- **e** Reliable due to interpolation and a strong correlation.

16

- **a** 0.67
- **b** Moderate positive linear relationship
- **c** $y = 0.21x - 2.09$
- **d** 15 computers
- **e** Very unreliable due to extrapolation and a moderate to weak correlation.

17

- **a** $r = -0.20$
- **b** $(1, 1.28)$
- **c** $r = 0.79$
- **d** Strong positive linear relationship
- **e** $y = 0.05x + 0.23$
- **f** 1.23
- **g** Despite a moderate correlation, unreliable due to extrapolation.